2008 F=ma Exam: Problem 19

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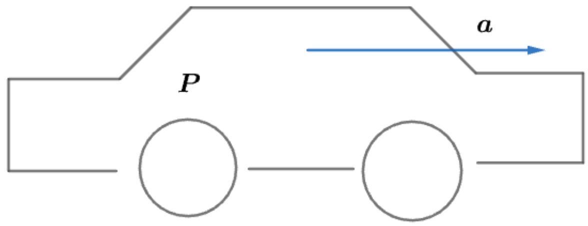

Since there is no energy loss, all power from the engine goes to kinetic energy,

$$E = Pt = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2Pt}{m}}$$

The instantaneous power is given by

$$P = Fv = mav$$

Substituting our expression for the velocity,

$$P = ma\sqrt{\frac{2Pt}{m}}$$

$$a \propto \frac{1}{\sqrt{t}}$$

If the time is doubled, the acceleration decreases by a factor of $\sqrt{2}$,

$$a = \frac{a_0}{\sqrt{2}}$$

so the answer is $\boxed{\text{B}}$.